

Secondary examination 2018

SCIENCE

Time: 3 1/4 hours

Total marks: 80

General instructions for the examinees.

1. Candidates must write first his or her roll number on the question paper compulsorily.
2. All questions are compulsory.
3. Write the Answer to each question in the given Answer book only.
4. For questions having more than one part the Answers to the other part should be written together in continuity.
5. If there is any error /difference /contradiction in Hindi and English versions of the paper the question of Hindi version should be treated valid.

6. Section	q.no	marks
A	1-11	1
B	12-21	3
C	22-27	4
D	28-30	5

7. There are internal services in question 27 to 30.

1. Write the name of enzyme secreted by salivary gland. [1]

Answer: Salivary amylase is secreted by salivary gland.

2. Write the name of Universal Donor blood group. [1]

Answer: Persons with O blood group are called Universal Donor.

3. Write general formula of Alkyne series. [1]

Answer: The general formula of Alkane series is C_nH_{2n} .

4. Write joule units in one kilowatt hour. [1]

Answer: 3×10^6 J

5. Manali Sanctuary is situated in which state. [1]

Answer: Manali Sanctuary is situated in Himachal Pradesh.

6. Write the name of one kharif crop. [1]

Answer; Rice is a very common kharif crop.

7. How many hotspots of biodiversity are there in the world? [1]

Answer: There are 34 biosphere hotspot Biodiversity in the world.

8. Write the name of instrument which is used to measure the blood pressure. [1]

Answer: Sphygmomanometer is the instrument which is used to measure the blood pressure.

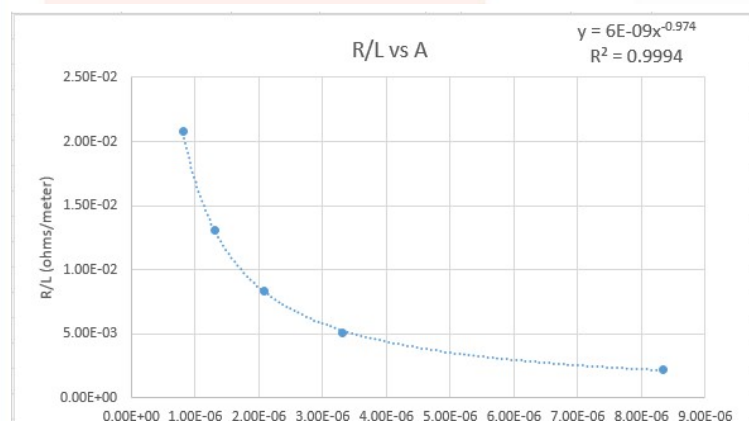
9. Write the name of an antibody present in the mother's milk. [1]

Answer: IgA is present in mother's milk.

10. Which vaccine is used in the treatment of erythroblastosis foetalis? [1]

Answer: Rhogam is used in the treatment of erythroblastosis foetalis.

11. Draw graph between area of cross section a and resistance of different conducting wires which are made by same material having equal length. [1]



SECTION - B

12. What are internal tectonic forces of earth .explain any two forces. [1 +2= 3]

Answer: In great Himalaya regions many years back there existed the Tethis Ocean. Many forms of forces continuously work for changing the Earth surface. These forces are called tectonic forces. Tectonic forces are of two types. Internal moulding forces and external moulding forces.

- **Volcanoes-**In Internal moulding forces volcano is the most interesting and curious incident. In volcano there is movement inside the Earth and the surface begins shaking and a rupture is created in the crust of earth which allows hot Lava volcanic Ash and gases to escape from the Magma chamber below the surface of the earth. Sometimes a very heated rocks melt and flow out in the form of lava. This produces a very dangerous disaster which results in damage of life and material.
- **Tsunami-** It is one of the incidents which is produced by internal moulding forces. Due to Tsunami a high energy wave is generated in the sea. It is also known as seismic sea wave. These waves cause huge damage to the coastal areas the reason for generation of Tsunami is an earthquake of more than 7 magnitude which has originated at the sea bottom from the point of origin. The tsunami travelling towards the coastal areas bring rules losses and destruction. The debris and waste coming from Tsunami cause huge damage to human building animal.

13. What is Fossil? Write the name of two vestigial organs found in human body? [1+1+1=3].

Answer: A Fossil is a naturally preserved remains or traces of animals and plants that lived in in geological past fossils are formed when minerals in groundwater replace materials in bones and tissues creating a replica in stone of original organism or if it's trace.

Two vestigial organs in human body are wisdom teeth and tailbone or coccyx.

14. Write the name of first spacecraft of India. Explain the importance of satellites launched by India. [1 +2=3]

Answer: Aryabhata was the first satellite launched by India.

Satellite are gathering information regarding the climate acceptor and sending it. They are providing assistance in the successful operation of radio, television, telephone, internet telemedicine, distant education. India has gained measure success along with the development of a powerful rocket polar satellite launch vehicle PSLV. This help in launching a satellite into Sun synchronous orbit.

15. a. What quality of alcohol in body is required for punishment when measured by breath analyzer?

Answer: According to law 30mg alcohol is acceptable limit in hundred ml of blood. It is determined by breath analyzer. If more alcohol is found then it is punishable offence.

b. Explain the points of road safety education. .[1+2 =3]

Answer: Different points of road safety Education Act following:

1. Alcohol sedative medicine which slows down the mind process. If more alcohol is found while driving it is a punishable offence.
2. Good eye vision is important for safe driving.
3. Concave Mirrors are used to obtain parallel light beam in the vehicles it has to divert to see far objects. High intensity should be used at high National Highways so that the driver can see the far object from a proper distance.

16. a. Write any two diseases caused by virus. $\frac{1}{2}+1/2+1+1/2 +1/2=3$

Answer: Dengue and AIDS.

b. write the name of alkaloid found in tobacco.

Answer: Nicotine alkaloid is found in tobacco.

c. write two harmful effects of chewing tobacco.

Answer: The harmful effects of tobacco are as follow:

The possibility of Cancer increases many times in mouth tongue throat and lungs by regular use of tobacco. The nicotine available in tobacco increases the thickness of walls of arteries which increases the Heartbeat and blood pressure.

17. Write any one point difference between the following:

a. Positive and negative catalyst.

Positive catalyst	Negative catalyst
The catalyst which increase the rate of chemical reaction are called positive catalyst.	The catalyst that decreases the rate of chemical reaction are called negative catalyst.
Ex: $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$ In presence of catalyst MnO_2 and heat.	Ex: $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ In presence of catalyst glycerol.

(b) Thermolysis and electrolysis.

Thermolysis	Electrolysis
Thermal decomposition is the process of breaking down a chemical compound in the simpler substances by heating it.	Electrolysis is a process of forcing a chemical reaction that would not normally occur on its own it occurs by passing an electric current through the reactants.
Heat is necessary for the reaction.	Takes place by passing of electric current.

c. Addition and displacement reaction.

Addition reaction	Displacement reaction
The chemical reaction in which two or more than two reactants on combined together to form a single product is called addition reaction.	The chemical reaction in which atom or a group of atoms present in one reaction is replaced by atoms or group of atoms of other reactant is called displacement reaction.
.The formation of new bonds takes place between reactants.	Bonds present in reaction are broken and some new bonds are formed.

18. What is meant by jhum cultivation .write two main components of social forestry? [1+1+1=3]

Answer: Jhum farming is the main reason of the deforestation. In such type of farming the green stuff of a Special area is converted into ash by burning by which fertility of that land is increased and we can obtain good crops from this land for 2 years.

Social Forestry is done to acquire the goal of plantation on nearly 1 crore hectare of degraded land .It will result in increment in plantation and provide employment to people on large scale.

The main components of social Forestry are-

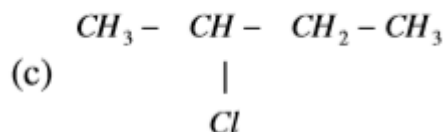
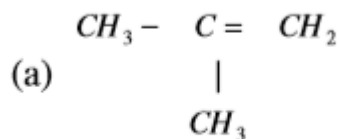
1. Agroforestry.
2. Plantation by the forest department on public spaces such as banks of canal, road, Hospital etc to Cater to community requirements.

19. Who is known as missile man .give the role of Dr. Panchanan Maheshwari in field of Botany. [1+2=3].

Answer: Dr. A. P. J. Abdul Kalam as known as the missile man.

Dr. Panchanan Maheshwari was a prominent Indian botanist. He did remarkable work on test tube fertilization of angiosperms. This has allowed the creation of a new hybrid plants that could not previously be cross breed naturally.

20. Write IUPAC name of following compounds. [1+1+1=3]



a) 2-Methane-Propene

b) Butene

c) 2-chloro-Butane

21. What is waste? Explain two method of waste management. [1+1+1=3].

Answer: Useless materials and products which are obtained at the end of any process is known as wastes. It refers to those material which are utilized and throw away assuming them to be useless. There are different methods of waste management depending on the types of waste substance.

Landfill -landfills are used usually formed in mines which are not in use mining blocked, regions etc. This is extremely clean and relatively less expensive method of waste disposal and it is commonly practiced in most of the countries.

Incineration -In this process of waste disposal waste material is burnt. Due to this the waste get transform into heat gas vapour and ash.

22a. Write the name of silkworm larva.

Answer: The silkworm of larva is called caterpillar.

b. write the two products of Apiculture .

Answer: Honey and Bee wax are the two products of Apiculture.

c. How the thread of silk is made by silkworm. $1+1/2+1/2+1/2+2=4$].

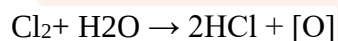
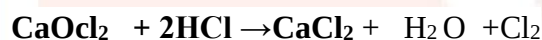
Answer: There are one pair of salivary glands in larva of silk s which are known as Silk gland. Secretion is available in liquid form. When it comes in contact with air it becomes hard. A fully developed larvae stop eating and start making cocoon. It covers itself completely with silk threads. The inactive larvae is known as puma. It is made of approximately 1000 to 1200 meter long silk thread.

23. What is reciprocal cross? Explain the phenotypic and genotypic ratio obtained in offspring's when F1 generation is cross with homozygous dominant parents. [1+3=4]

Answer: The cross or the hybridization in which plant ‘‘A’’ (TT) is used as male parents while plant ‘‘B’’ tt is used as female parents. In the second case plant ‘‘A’’ is used as female parents and plant ‘‘B’’ (tt) is used as male parent which is called reciprocal cross.

24. a. Write formula of bleaching powder. Explain its bleaching process.

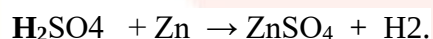
Answer: The chemical name of bleaching powder is calcium oxychloride and formula is CaOCl_2 . Is yellow solid substance having characteristic odour. It gives chlorine gas on reaction with dilute acid. Free chlorine gas is liberated from bleaching powder forms nascent oxygen (O) on combination with water .The oxygen performs bleaching action and acts as an oxidant.



Coloured substance + [O] $\rightarrow$ Colorless substance.

b. Draw labeled diagram of chemical reaction of Zinc metal with dilute H_2SO_4 .

When zinc powder is added to a solution of sulphuric acid zinc sulphate and hydrogen gas produced. Hydrogen can be observed effervescing from the solution. The chemical equation for this reaction is as follows:



25. A. If force is applied upon a body of 40 kg then its velocity increases from one meter/second 22 meter /second. Determined work done by the force.

Answer: Given-

Mass= 40Kg

Displacement= (22m/s- 1m/s)

F= Mg

$$= 40 \times 10 = 400\text{N}$$

$$\text{Energy} = F \times d$$

$$= 400 \times 21$$

$$= 8400\text{J}$$

b) A spring of constant K is equal to 4 into 10 to the power 3 Newton per meter is compressed by two centimeter then determine potential energy stored in the spring. [2+2=4]

$$\text{Constant} = 4 \times 10^3$$

$$\text{Compression}(x) = 2 \times 0.01 = 0.02\text{m}$$

$$\text{Potential energy} = \frac{1}{2} Kx^2$$

$$= \frac{1}{2} \times 4 \times 10^3 \times 0.02 \times 0.02$$

$$= 0.04 \times 0.02$$

$$= 0.0008$$

$$= 0.8 \text{ J}$$

26. What is biodiversity? Explain the in situ and ex situ conservation. 2+2=4

Biodiversity can be defined as the variety and variability among living organism and ecological complexes in which they occur. It ranges from plant that is lichens to giant banyan and Redwood trees and from very small bacteria and giant whale and elephant. Biodiversity conservation can be done by following two ways:

In situ Conservation -The natural habitats are best suitable for development of birth and growth of organisms Conservation which can be done in natural habitat is known as in situ conservation.

Ex -situ conservation- In this method of conservation endangered species are shifted from their Natural Habitat to artificial Habitat for reservation .For conservation of plant species Botanical Garden, seed Bank tissue culture laboratories has been established

OR

What is Genetic diversity? Explain two reasons of threat to biodiversity?

[2+2=4]

The difference or variation between a species due to genetic makeup of a species is known as Genetic diversity. This diversity is found between the population of a group of species or around the various members of its species.

The two reasons for threat to biodiversity are following.

1. Habitat loss - We are destroying the natural habitat to fulfill the requirement of increased population and building houses for human and agricultural expansion.

2.,Habitat fragmentation- Habitats of wild animals which where is spread in long area are now destructed to construction of roads tracks, railroad track, gas pipelines, Canal, electric lines, dams, etc. It affect the activity of wildlife and they feel unsafe from these activities. Animals lost their life due to accident with vehicles.

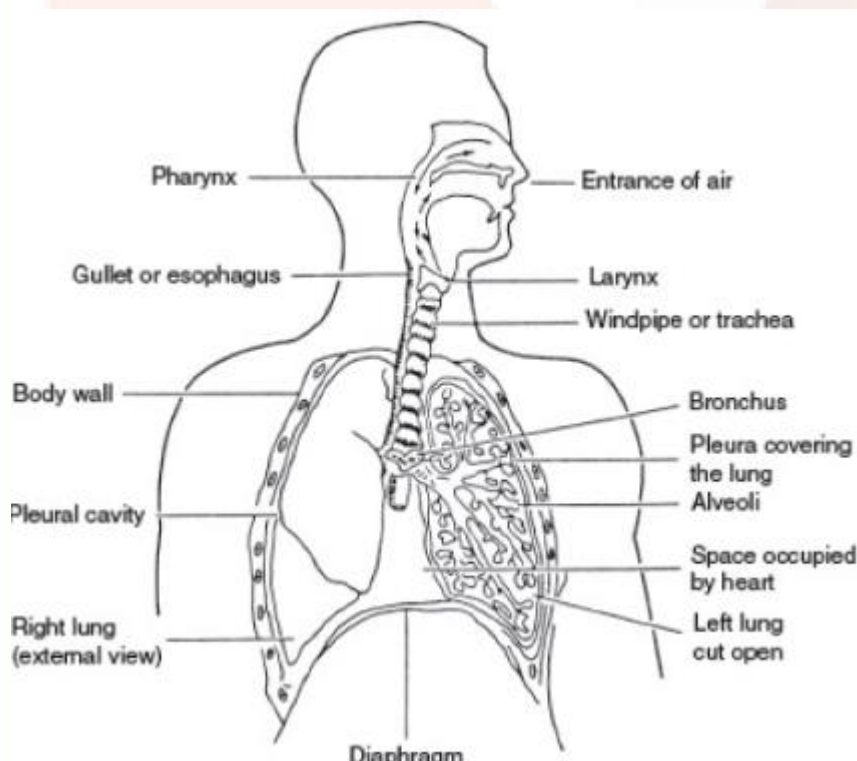
SECTION – D

28. a. What is respiration?

1+2+2=5

Answer: Respiration is the oxidation of nutrients in living cells to release energy for biological works it takes place in all the cells of the body. It also produces carbon dioxide as toxic substances which is eliminated from the body.

b. Draw a labelled diagram of human respiratory system.



RESPOIRATORY SYSTEM

c. Explain the mechanism of respiration.

The process of respiration takes place in two step.

External respiration- This is the exchange of gases between alveoli and blood. The exchange of gases between alveoli and pulmonary capillaries is called external respiration. Alveoli are the primary sites of gas exchange. O_2 and CO_2 are exchange in this sites by simple diffusion based on pressure/concentration gradients, solubility of gas and thickness of membrane, surface area of respiratory membrane.

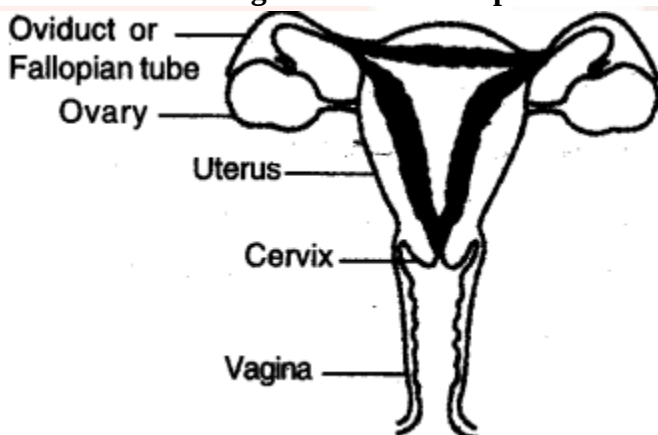
Internal respiration- This is the exchange of gases between blood and tissue cells. Exchange of gases between tissue blood capillaries and tissue cells is called internal respiration .it takes place by simple diffusion process.

OR

a. Write the name of primary sex organ in female.

Answer: Primary sex organs in female include a pair of ovaries, accessory ducts and external genitalia.

b. Draw a labelled diagram of female reproductive system.



FEMALE REPRODUCTIVE SYSTEM

c. Explain two phases of human reproduction [1+2+2=5]

Answer: The phases of human reproduction are following.

Gametogenesis - It is the process of formation of differentiation of haploid gametes sperm and ova. They are formed from diploid primary germ cells present in primary sex organs called gonads. Gonads are testis in male and ovaries in female respectively.

Fertilization - it is the most important event in reproduction. It is the fusion of male and female gametes to form a diploid zygote. It is also known as syngamy.

29. a. Elements of which block shows variable valency in periodic table. [1+2+2=5]

Answer: d- block elements lanthanides and actinides represent more than one valency. It is called variable valency.

b. Why size of anion is greater than its atom?

Answer: The number of electrons in the outermost shell increases during the formation of an anion and the value of effective nuclear charge decreases so the size of an anion is always greater than the neutral atoms.

c. CaH_2NaH , SiH_4 , AlH_3 .

Give the valency is of Ca, Na, Si, Al in the above compounds.

Answer: Valency of Ca is 2, Na is 1. Si is 4, Al is 3

OR

a) Write the name or any two metalloids.

Answer: Silicon Arsenic are metalloids.

b. How does atomic size changes on going from left to right in a period .Explain with reason.

Answer: Atomic size decreases on moving from left to right in a period due to number of in the Shell and increase in effective nuclear charge.

c. Arrange the following elements in ascending order of their atomic size. Na, Cs, Li, K.

Answer: Li, Na, K, Cs

30. a. The sun is visible a little before the actual sunrise and a little after the actual sunset explain the reason. 1+1+1+2=5

Answer: The sun is visible to us two minutes before the sunrise and two minutes after the sunset because of the bending of light due to atmospheric refraction. The light from the sun got reflected by the atmospheric layer and reaches the observer this causes the observer to think that light is coming straight from the Sun.

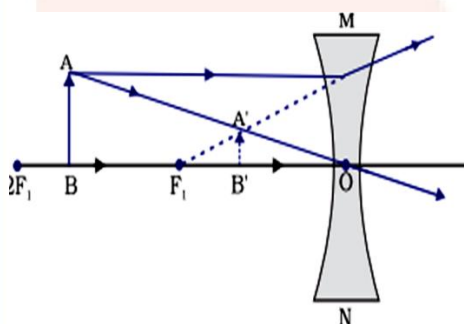
b. What does mean by Dispersion of white light.

Answer: The phenomenon of splitting of white light into its constituent colour when it passes through a prism is called is dispersion. This is band of seven colours obtained the VIBGYOR (V= violet, I=Indigo, B= blue, G= Green, Y= Yellow, O= Orange and R= Red).

c. What is mean by total internal reflection of light?

Answer: When their light travelling from denser medium to rarer medium the ray of light bends away from the normal after the refraction. If the angle of incidence increases the angle of refraction also increases. The special angle of incidence at which angle of refraction is 90° is called critical angle. If we increase angle of incidence then the Ray of light comes back into the same design media for total internal reflection.

d. Draw a ray diagram for image formed by concave lens when the object is placed between infinity and optical Centre O.



OR

A. Pencil partially immersed in water in a glass tumbler appears skewed.why?

Answer: Pencil partially immersed in water in a glass tumbler appears skewed due to refraction of light. For the part of pencil which is immersed in water light changes its medium from air to glass and back to air. The refraction in and out of glass causes little deviation in the original path of light.

b. What does mean by power of lens?

Answer: The ability of lens to converge or diverge light rays falling on it is called power of lens. The ray is more converged after refraction in convex lens of small focal length and focused close to the lens. On the contrary the rays are minimum converged in convex lens of focal length

and focused at large distance. When the light rays are incident on concave lens then the rays are diverted less or more according to the focal length.

c. What is astigmatism of human eyes?

Answer: Astigmatism is due to the irregularities developed in the cornea of the eye. In such type of defected eye cornea is not perfectly spherical and hence, a person cannot focus his eyes on both horizontal and vertical planes simultaneously. This type of defect is corrected using spectacles fitted with a suitable cylindrical lens.

d. Draw a ray diagram of image formed by concave mirror when object is placed between centre of curvature 'C' and focus 'F'.

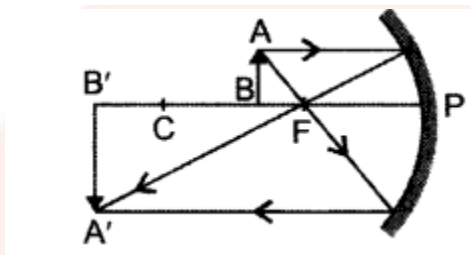


Image formed is beyond the center of curvature. Real, inverted and bigger in size than the object.